02477 Bayesian ML — Distributions & Conjugate Models

Functional forms · moments · conjugate posterior update rules

Discrete Continuous Multivariate
Conjugate GP

Discrete Continuous Multivariate Conjugate Models Gaussian Process Summary Table

Discrete Distributions

Bernoulli
$\mathrm{Ber}(\pi)$

Support	$y \in \{0, 1\}$
Params	$\pi \in (0, 1)$
PMF	$p(y\|\pi) = \pi^y (1-\pi)^{1-y}$
Mean	π
Variance	$\pi(1-\pi)$
Used for	Binary classification likelihoods; GP classification

Binomial
$\mathrm{Bin}(n, \pi)$

Support	$k \in \{0, 1, \dots, n\}$
Params	$n \in \mathbb{N},\ \pi \in (0, 1)$
PMF	$p(k\|n, \pi) = \binom{n}{k} \pi^k (1-\pi)^{n-k}$
Mean	$n\pi$
Variance	$n\pi(1-\pi)$
Used for	Count of successes; Beta-Binomial (Week 1)

Poisson
$\mathrm{Pois}(\lambda)$

Support	$k \in \{0, 1, 2, \dots\}$
Params	$\lambda > 0$
PMF	$p(k\|\lambda) = \dfrac{\lambda^k e^{-\lambda}}{k!}$
Mean	λ
Variance	λ

Used for	Count data; Bayesian Poisson regression (Week 8)

Categorical $\mathrm{Cat}(\boldsymbol{\pi})$

Support	$y \in \{1, \ldots, K\}$
Params	$\boldsymbol{\pi} \in \Delta^{K-1},\ \sum_k \pi_k = 1$
PMF	$p(y = k \mid \boldsymbol{\pi}) = \pi_k$
Mean	π_k for class k
Used for	Multi-class labels; GMM assignments (Weeks 7, 10)

Continuous Univariate Distributions

Beta $\mathrm{Beta}(a, b)$

Support	$\pi \in (0, 1)$
Params	$a, b > 0$ (shape parameters)
PDF	$\frac{\pi^{a-1}(1-\pi)^{b-1}}{B(a,b)}, \quad B(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$
Mean	$\frac{a}{a+b}$
Variance	$\frac{ab}{(a+b)^2(a+b+1)}$
Mode	$\frac{a-1}{a+b-2}$ for $a, b > 1$
Used for	Prior/posterior over Bernoulli/Binomial π (Week 1)

Gaussian (univariate) $\mathcal{N}(\mu, \sigma^2)$

Support	$x \in \mathbb{R}$
Params	$\mu \in \mathbb{R},\ \sigma^2 > 0$
PDF	$\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$
Mean	μ
Variance	σ^2
Entropy	$\frac{1}{2} \ln(2\pi e\, \sigma^2)$
Used for	Noise, priors, posteriors, likelihoods — ubiquitous

Gamma $\mathrm{Gamma}(a, b)$

Support	$x > 0$
Params	$a > 0$ (shape), $b > 0$ (rate = 1/scale)
PDF	$\frac{b^a}{\Gamma(a)} x^{a-1} e^{-bx}$
Mean	a/b
Variance	a/b^2
Mode	$(a-1)/b$ for $a \geq 1$
Used for	Prior over Poisson rate λ or precision τ (Weeks 8, 10)
Note	scipy uses `scale=1/b` not rate

Student-t $\mathcal{T}(\mu, \sigma^2, \nu)$

Support	$x \in \mathbb{R}$
Params	μ (loc), σ^2 (scale), $\nu > 0$ (dof)
PDF	$p(x) \propto \left(1 + \frac{(x-\mu)^2}{\nu\sigma^2}\right)^{-(\nu+1)/2}$
Mean	μ (for $\nu > 1$)
Variance	$\nu\sigma^2/(\nu-2)$ for $\nu > 2$
Used for	Marginal likelihood; robust regression likelihood (Week 11)
Origin	Arises marginalising σ^2 out of $\mathcal{N}(\mu, \sigma^2)$ with Inverse-Gamma prior

Multivariate Distributions

Multivariate Gaussian $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$

Support	$\mathbf{x} \in \mathbb{R}^D$
Params	$\boldsymbol{\mu} \in \mathbb{R}^D$, $\boldsymbol{\Sigma} \succ 0$
PDF	$(2\pi)^{-D/2}\lvert\boldsymbol{\Sigma}\rvert^{-1/2} \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right)$
Mean	$\boldsymbol{\mu}$
Covariance	$\boldsymbol{\Sigma}$
Entropy	$\frac{1}{2}\ln\left((2\pi e)^D \lvert\boldsymbol{\Sigma}\rvert\right)$
Key props	Marginals, conditionals, and affine transforms are all Gaussian

Dirichlet $\mathrm{Dir}(\boldsymbol{\alpha})$

Support $\boldsymbol{\pi} \in \Delta^{K-1}$ (simplex, $\sum_k \pi_k = 1$)

Params $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_K),\ \alpha_k > 0$

PDF $\dfrac{\Gamma(\alpha_0)}{\prod_k \Gamma(\alpha_k)} \prod_k \pi_k^{\alpha_k - 1}, \quad \alpha_0 = \sum_k \alpha_k$

Mean $\mathbb{E}[\pi_k] = \alpha_k / \alpha_0$

Variance $\dfrac{\alpha_k(\alpha_0 - \alpha_k)}{\alpha_0^2(\alpha_0 + 1)}$

Used for Prior over mixture weights; conjugate to Categorical (Week 10)

Note $\alpha_k = 1$ = Uniform; larger α_k concentrates mass on π_k

Wishart $\mathcal{W}(\nu, \mathbf{V})$

Support $\boldsymbol{\Lambda} \succ 0$ ($D \times D$ positive definite)

Params $\nu > D - 1$ (dof), $\mathbf{V} \succ 0$ (scale matrix)

PDF $p(\boldsymbol{\Lambda}) \propto |\boldsymbol{\Lambda}|^{(\nu - D - 1)/2} \exp\left(-\frac{1}{2}\mathrm{tr}(\mathbf{V}^{-1}\boldsymbol{\Lambda})\right)$

Mean $\nu \mathbf{V}$

Mode $(\nu - D - 1)\mathbf{V}$ for $\nu > D + 1$

Used for Prior over precision matrices $\boldsymbol{\Lambda} = \boldsymbol{\Sigma}^{-1}$ in GMM (Week 10)

Mean-field Gaussian family $\prod_i \mathcal{N}(m_i, v_i)$

Form $q(\mathbf{w}) = \prod_{i=1}^{D} \mathcal{N}(w_i | m_i, v_i)$

Cov Diagonal — off-diagonals forced to 0

Entropy $H[q] = \sum_{i=1}^{D} \frac{1}{2} \ln(2\pi e\, v_i)$

Moments $\mathbb{E}[w_i] = m_i$

$\mathbb{E}[w_i^2] = m_i^2 + v_i$

$\mathbb{E}[w_i w_j] = m_i m_j$ (by independence)

Used for VI in Weeks 10–11; always underestimates posterior variance

Conjugate Models — Posterior Update Rules

Beta–Binomial Week 1

$\pi \sim \mathrm{Beta}(a, b)$ + $k|\pi \sim \mathrm{Bin}(n, \pi)$ → $\pi|k \sim \mathrm{Beta}(a', b')$

POSTERIOR UPDATE

$a' = a + k \qquad b' = b + n - k$

POSTERIOR PREDICTIVE

$p(k^*|k) = \mathrm{BetaBin}(n, a', b')$

Posterior mean: $\frac{a+k}{a+b+n}$. Prior counts: a pseudo-successes, b pseudo-failures.

Gaussian–Gaussian (unknown mean)

Week 3

$\mu \sim \mathcal{N}(\mu_0, \tau^2)$ + $y_n|\mu \overset{iid}{\sim} \mathcal{N}(\mu, \sigma^2)$ → $\mu|\mathbf{y} \sim \mathcal{N}(m, v)$

POSTERIOR UPDATE

$$v = \left(\frac{1}{\tau^2} + \frac{N}{\sigma^2}\right)^{-1}, \qquad m = v\left(\frac{\mu_0}{\tau^2} + \frac{\sum_n y_n}{\sigma^2}\right)$$

POSTERIOR PREDICTIVE

$p(y^*|\mathbf{y}) = \mathcal{N}(y^* \mid m,\ v + \sigma^2)$

Bayesian Linear Regression

Week 3

$\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \alpha^{-1}I)$ + $\mathbf{y}|\mathbf{w} \sim \mathcal{N}(\Phi\mathbf{w}, \sigma^2 I)$ → $\mathbf{w}|\mathbf{y} \sim \mathcal{N}(\mathbf{m}, S)$

POSTERIOR UPDATE

$$S^{-1} = \alpha I + \tfrac{1}{\sigma^2}\Phi^\top\Phi, \qquad \mathbf{m} = \tfrac{1}{\sigma^2} S\,\Phi^\top\mathbf{y}$$

POSTERIOR PREDICTIVE

$p(y^*|\mathbf{y}, x^*) = \mathcal{N}(y^* \mid \mathbf{m}^\top\boldsymbol{\phi}^*,\ \boldsymbol{\phi}^{*\top} S\boldsymbol{\phi}^* + \sigma^2)$

Marginal likelihood: $p(\mathbf{y}|\alpha, \sigma^2) = \mathcal{N}(\mathbf{y} \mid \mathbf{0},\ \sigma^2 I + \alpha^{-1}\Phi\Phi^\top)$

Gamma–Poisson

Week 8

$\lambda \sim \mathrm{Gamma}(a, b)$ + $k_n|\lambda \overset{iid}{\sim} \mathrm{Pois}(\lambda)$ → $\lambda|\mathbf{k} \sim \mathrm{Gamma}(a', b')$

POSTERIOR UPDATE

$a' = a + \sum_n k_n \qquad b' = b + N$

POSTERIOR PREDICTIVE

Negative Binomial: $p(k^*|\mathbf{k}) = \mathrm{NegBin}(a', b'/(b'+1))$

Posterior mean of λ: a'/b'. Used in mortality/count regression (Week 8).

Dirichlet–Categorical

Week 10

$\boldsymbol{\pi} \sim \mathrm{Dir}(\boldsymbol{\alpha})$ + $y_n|\boldsymbol{\pi} \overset{iid}{\sim} \mathrm{Cat}(\boldsymbol{\pi})$ → $\boldsymbol{\pi}|\mathbf{y} \sim \mathrm{Dir}(\boldsymbol{\alpha}')$

POSTERIOR UPDATE

$\alpha'_k = \alpha_k + N_k \qquad N_k = \sum_n \mathbf{1}[y_n = k]$

POSTERIOR PREDICTIVE

$p(y^* = k|\mathbf{y}) = \alpha'_k/\alpha'_0 \quad \text{where } \alpha'_0 = \sum_k \alpha'_k$

Normal–Gamma (joint prior on μ, λ)

Week 10 VI

$\mu|\lambda \sim \mathcal{N}(m_0, (\beta_0\lambda)^{-1}),\ \lambda \sim \mathrm{Gamma}(a_0, b_0)$

$y_n|\mu,\lambda \overset{iid}{\sim} \mathcal{N}(\mu, \lambda^{-1})$ → Normal-Gamma posterior

POSTERIOR UPDATE (N OBSERVATIONS, $\bar{Y} = \frac{1}{N}\sum Y_N$)

$$\beta_N = \beta_0 + N, \quad m_N = \frac{\beta_0 m_0 + N\bar{y}}{\beta_N}, \quad a_N = a_0 + \frac{N}{2}$$

$$b_N = b_0 + \frac{1}{2}\left[\sum_n (y_n - \bar{y})^2 + \frac{\beta_0 N(\bar{y} - m_0)^2}{\beta_N}\right]$$

MARGINAL PREDICTIVE (INTEGRATING OUT Μ, Λ)

$p(y^*|\mathbf{y}) = \mathcal{T}\left(y^* \mid m_N,\ \frac{b_N(\beta_N+1)}{a_N\beta_N},\ 2a_N\right)$

Normal–Wishart (GMM clusters)

Week 10 GMM

$\boldsymbol{\Lambda} \sim \mathcal{W}(\nu_0, \mathbf{V}_0),\ \boldsymbol{\mu}|\boldsymbol{\Lambda} \sim \mathcal{N}(\mathbf{m}_0, (\beta_0\boldsymbol{\Lambda})^{-1})$

$\mathbf{y}_n|\boldsymbol{\mu}, \boldsymbol{\Lambda} \overset{iid}{\sim} \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Lambda}^{-1})$ → Normal-Wishart posterior

POSTERIOR UPDATE

$$\beta_N = \beta_0 + N, \quad \mathbf{m}_N = \frac{\beta_0 \mathbf{m}_0 + N\bar{\mathbf{y}}}{\beta_N}, \quad \nu_N = \nu_0 + N$$

$$\mathbf{V}_N^{-1} = \mathbf{V}_0^{-1} + \sum_n (\mathbf{y}_n - \bar{\mathbf{y}})(\mathbf{y}_n - \bar{\mathbf{y}})^\top + \tfrac{\beta_0 N}{\beta_N}(\bar{\mathbf{y}} - \mathbf{m}_0)(\bar{\mathbf{y}} - \mathbf{m}_0)^\top$$

One Normal-Wishart per cluster in Bayesian GMM CAVI. CAVI alternates updating cluster params and responsibilities.

Laplace Approximation (non-conjugate)

Weeks 4, 6, 7

For non-conjugate models (logistic regression, GP classification) where the posterior is intractable.

Any $p(w)$ + Non-Gaussian $p(\mathbf{y}|w)$ → $p(w|\mathbf{y}) \approx \mathcal{N}(w|\hat{w}, S)$

CONSTRUCTION

$$\hat{w} = \arg\max_w \log p(w, \mathbf{y}) \qquad S^{-1} = -\nabla^2 \log p(w, \mathbf{y})\big|_{\hat{w}}$$

PROBIT APPROXIMATION OF POSTERIOR PREDICTIVE

$$p(y^* = 1|\mathbf{y}, x^*) \approx \Phi\left(\frac{\mu^*}{\sqrt{8/\pi + \sigma^{2*}}}\right)$$

μ^*, σ^{2*} = mean/variance of $f^* = w^\top \phi(x^*)$ under the Laplace approx

Gaussian Process

GP Prior + Gaussian Likelihood → GP Posterior (Regression)

$f \sim \mathcal{GP}(0, k)$ + $\mathbf{y}|f \sim \mathcal{N}(f, \sigma^2 I)$ → $f^*|\mathbf{y} \sim \mathcal{N}(\mu^*, \sigma^{2*})$

POSTERIOR AT X^*

$$\mu^* = \mathbf{k}^{*\top}(K + \sigma^2 I)^{-1}\mathbf{y}$$

$$\sigma^{2*} = k(x^*, x^*) - \mathbf{k}^{*\top}(K + \sigma^2 I)^{-1}\mathbf{k}^*$$

GP Classification (Laplace) + Kernel Properties

$f \sim \mathcal{GP}(0, k)$ + $y_n|f_n \sim \mathrm{Ber}(\sigma(f_n))$ → $p(f|\mathbf{y}) \approx \mathcal{N}(f|\mathbf{m}, S)$

POSTERIOR AT X^* GIVEN LAPLACE APPROX $(\mathbf{M}, S)$

$$\mu_{f^*} = \mathbf{k}^{*\top} K^{-1}\mathbf{m}, \quad \sigma^2_{f^*} = k(x^*, x^*) - \mathbf{k}^{*\top} K^{-1}$$

Predictive:

$\mathbf{k}^*$ = covariance vector between x^* and all training points; $K_{ij} = k(x_i, x_j)$

KEY DISTRIBUTIONS

Prior predictive:
$p(y^*|x^*) = \mathcal{N}(0,\ k(x^*, x^*) + \sigma^2)$
Post. predictive:
$p(y^*|\mathbf{y}, x^*) = \mathcal{N}(\mu^*,\ \sigma^{2*} + \sigma^2)$
Marginal likelihood:
$p(\mathbf{y}) = \mathcal{N}(\mathbf{y} \mid \mathbf{0},\ K + \sigma^2 I)$

$p(y^* = 1|\mathbf{y}, x^*) \approx \Phi\left(\mu_{f^*}/\sqrt{8/\pi + \sigma^2_{f^*}}\right)$

COMMON KERNELS

SE: $\kappa^2 \exp\left(-\|x - x'\|^2/2\ell^2\right)$ — stationary, isotropic, C^∞

Matérn 1/2: $\kappa^2 \exp(-\|x - x'\|/\ell)$ — stationary, C^0

Matérn 3/2: $\kappa^2(1 + \sqrt{3}\|r\|/\ell)\exp(-\sqrt{3}\|r\|/\ell) - C^1$

Linear: $k(x, x') = xx'$ — NOT stationary (breaks on xx' term)

Stationary iff $k(x, x') = f(x - x')$. Isotropic iff stationary AND $k = f(\|x - x'\|)$.

All-in-One Summary Table

DISTRIBUTION	PARAMS	MEAN	VARIANCE	CONJUGATE PRIOR FOR	POSTERIOR UPDATE R
Beta(a, b)	$a, b > 0$	$\frac{a}{a+b}$	$\frac{ab}{(a+b)^2(a+b+1)}$	Binomial π	$a' = a + k,\ b' = b + r$
Gamma(a, b)	$a, b > 0$	a/b	a/b^2	Poisson rate λ	$a' = a + \sum k_n,\ b' = i$
Dirichlet$(\boldsymbol{\alpha})$	$\alpha_k > 0$	α_k/α_0	$\frac{\alpha_k(\alpha_0-\alpha_k)}{\alpha_0^2(\alpha_0+1)}$	Categorical $\boldsymbol{\pi}$	$\alpha'_k = \alpha_k + N_k$
Gaussian(μ, σ^2)	$\mu \in \mathbb{R}, \sigma^2 > 0$	μ	σ^2	Gaussian mean (known σ^2)	$v = (1/\tau^2 + N/\sigma^2)^{-1}$
MVN$(\boldsymbol{\mu}, \boldsymbol{\Sigma})$	$\boldsymbol{\mu}, \boldsymbol{\Sigma} \succ 0$	$\boldsymbol{\mu}$	$\boldsymbol{\Sigma}$	Linear model weights $\mathbf{w}$	$S^{-1} = \alpha I + \frac{1}{\sigma^2}\Phi^\top\Phi,$
Wishart$(\nu, \mathbf{V})$	$\nu > D - 1, \mathbf{V} \succ 0$	$\nu\mathbf{V}$	—	MVN precision $\boldsymbol{\Lambda}$	$\nu' = \nu + N$; $\mathbf{V}'$ updat
Student-t(μ, σ^2, ν)	$\mu, \sigma^2 > 0, \nu > 0$	μ ($\nu > 1$)	$\frac{\nu\sigma^2}{\nu-2}$ ($\nu > 2$)	—	Arises marginalising σ^2 prior
GP$(0, k)$	kernel k	$\mathbf{0}$	K (kernel matrix)	Regression f (with Gaussian lik.)	$\mu^* = \mathbf{k}^\top(K + \sigma^2 I)^{-1}$

QUICK CODE — SCIPY.STATS

```
from scipy.stats import (
    bernoulli, binom, poisson, norm, multivariate_normal as mvn,
    beta, gamma, dirichlet, t as student_t
)
# Note: scipy Gamma uses scale=1/rate, not rate directly
gamma.pdf(x, a=3, scale=1/b)           # Gamma(a, b=rate)
norm.interval(0.90, loc=mu, scale=sigma)  # 90% CI
norm.interval(0.95, loc=mu, scale=sigma)  # 95% CI
mvn.logpdf(y, mean=np.zeros(N), cov=C)   # log N(y|0,C)
norm.cdf(mu_f / np.sqrt(8/np.pi + var_f))  # probit approx
```